

Selenium

Web Application Testing

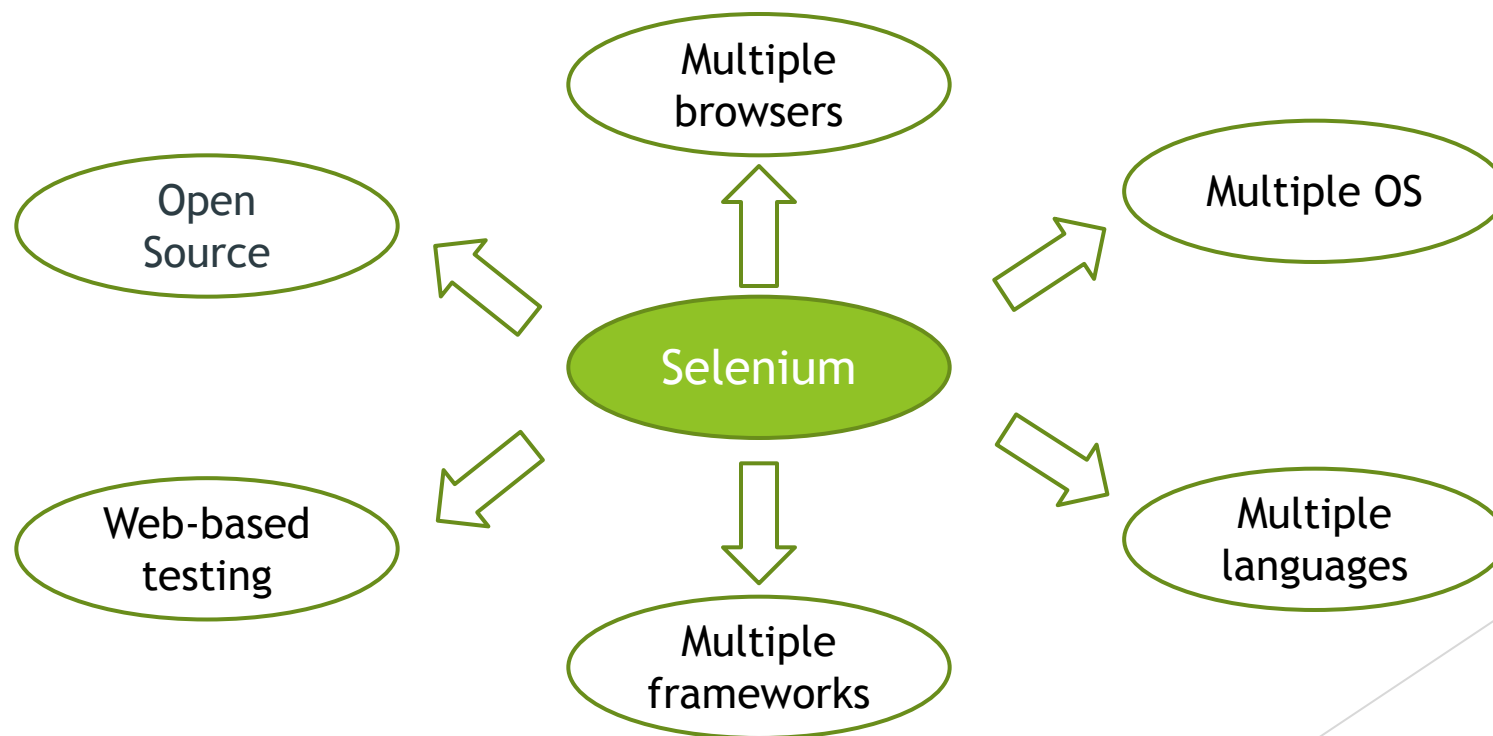
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What is Selenium?



What is Selenium?

- ▶ Framework for automating web applications/browsers
 - ▶ “Boring” web-based tasks
 - ▶ Testing web applications



What is Selenium?

- ▶ Framework for automating web applications/browsers
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What is Selenium IDE

- ▶ IDE plugin for Mozilla Firefox (www.mozilla.org)
- ▶ Record and playback user actions
- ▶ Very simple to use
- ▶ Difficult to use in more advanced tests

- Easy record and play back
- Capable of converting tests in html, Java, C# and various other languages
- No prior programming language experience is required
- Logging capabilities using file logging plug-in
- Debug and set breakpoints
- Flexibility & Extensibility

- A firefox plug -in, thus its support is limited to firefox only
- Doesn't supports iterations and conditional statements
- Doesn't supports error handling
- Doesn't supports test script dependency or grouping
- Doesn't supports database testing

What is WebDriver

- ▶ Object oriented framework
- ▶ Can directly test various browsers
- ▶ Similar to Selenium RC but better
- ▶ Programming knowledge needed

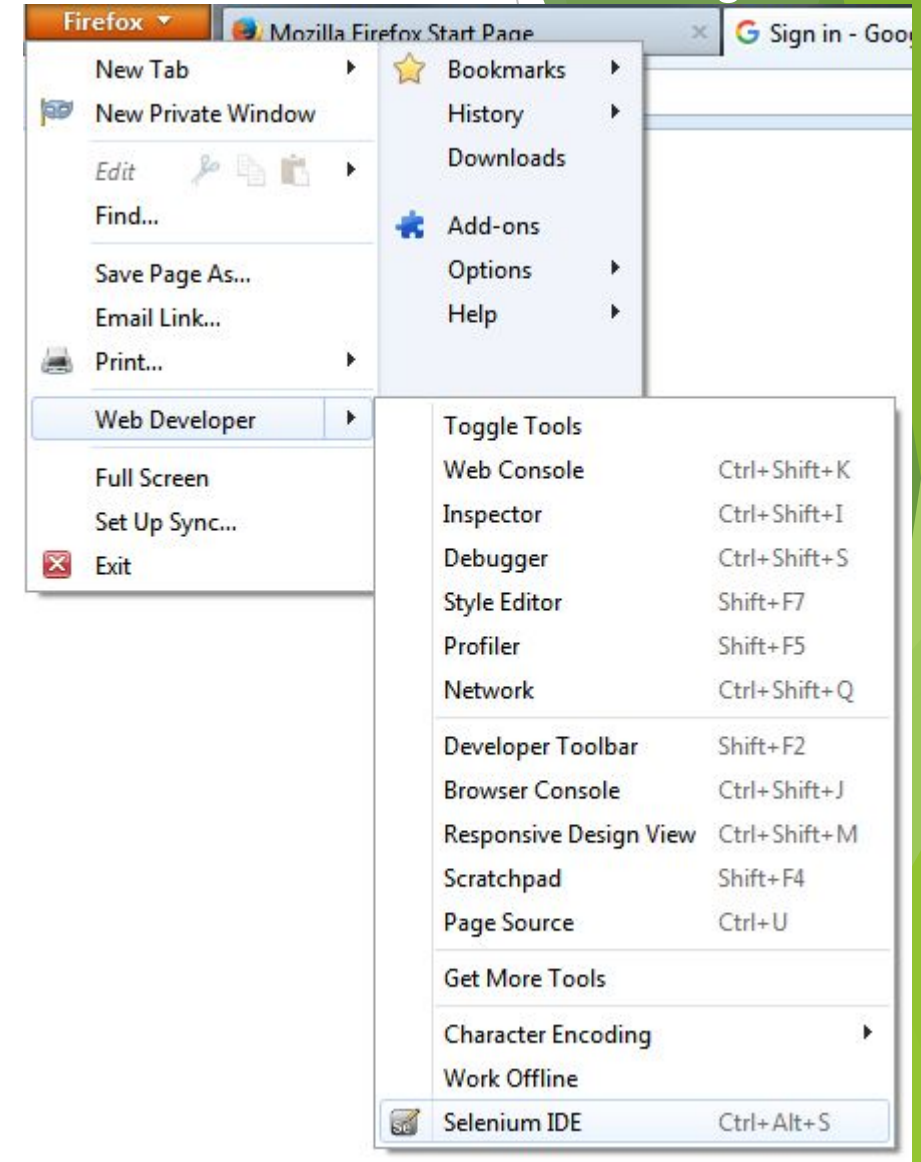
- Doesn't require selenium server to be started before executing test scripts
- Directly communicates with the browser
- A purely object oriented interface
- Supports dynamic finders
- Offers a wide range of utilities and classes that helps in handling alerts, navigations, Ajax calls & dropdowns
- Supports testing of WAP based applications(iphone/android)
- Support implementation of listeners
- Faster than Selenium RC

- A more complex API
- User is bound to have prior programming knowledge
- Doesn't supports mobile testing
- Migrating from selenium RC to WebDriver is a tiresome process
- Unable to test applications with flash/ flex objects

Selenium IDE

Selenium IDE install

- ▶ Step 1 - Open Firefox
- ▶ Step 2 - Download plugin: <http://tinyurl.com/cvjtm3m>
- ▶ Step 3 - Click install in the plugin manager
- ▶ Step 4 - Open: Firefox > Web Developer > Selenium IDE



Selenium IDE - Features

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The screenshot shows the Selenium IDE 2.9.0 interface. The following components are labeled with red lines:

- Menu Bar:** Located at the top, containing File, Edit, Actions, Options, and Help.
- Tool Bar:** Located below the menu bar, containing a speed slider (Fast/Slow), play, pause, and other control icons.
- Test Case Pane:** Located on the left side, showing a list of test cases: Untitled, Failed Testx, and Passed Testy.
- Address Bar:** Located at the top right, showing the Base URL: https://softwaretestinghelp.com.
- Test steps box:** A table in the center showing test steps. It has columns for Command, Target, and Value.
- Log, Reference, UI-Element & Rollup pane:** Located at the bottom, containing tabs for Log, Reference, UI-Element, and Rollup, along with Info and Clear buttons.

Command	Target	Value
open	/download/	
clickAndWait	link=Projects	
type	id=name	Shruti
clickAndWait	link=Selenium IDE	

Selenium IDE record example

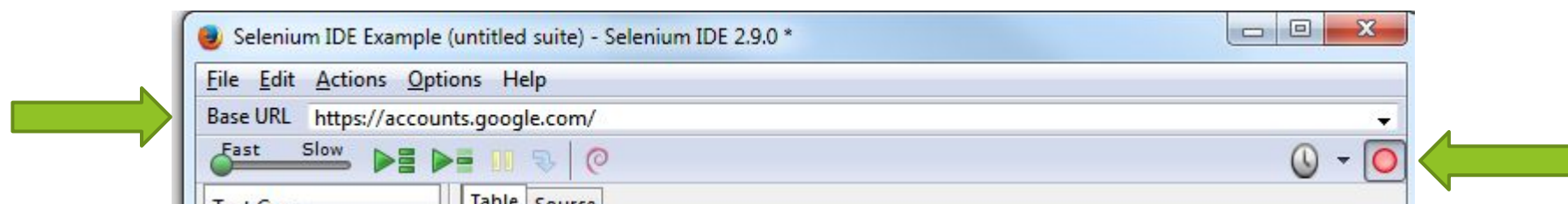
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Record a test to:

- ▶ 1 - Open “https://accounts.google.com”
- ▶ 2 - Verify you are redirected to the “Sign in - Google Accounts”
- ▶ 3 - Enter a valid username
- ▶ 4 - Enter a valid password
- ▶ 5 - Verify that you are redirected to the “My Account” page

Selenium IDE record example - Step 1

- ▶ Open Firefox and Selenium IDE
- ▶ Open page under test (“https://accounts.google.com”)
- ▶ Enter address (“https://accounts.google.com”) in Base URL in Selenium IDE
- ▶ Verify recording is “ON”



Selenium IDE record example - Step 2

- Right click anywhere not an image or link

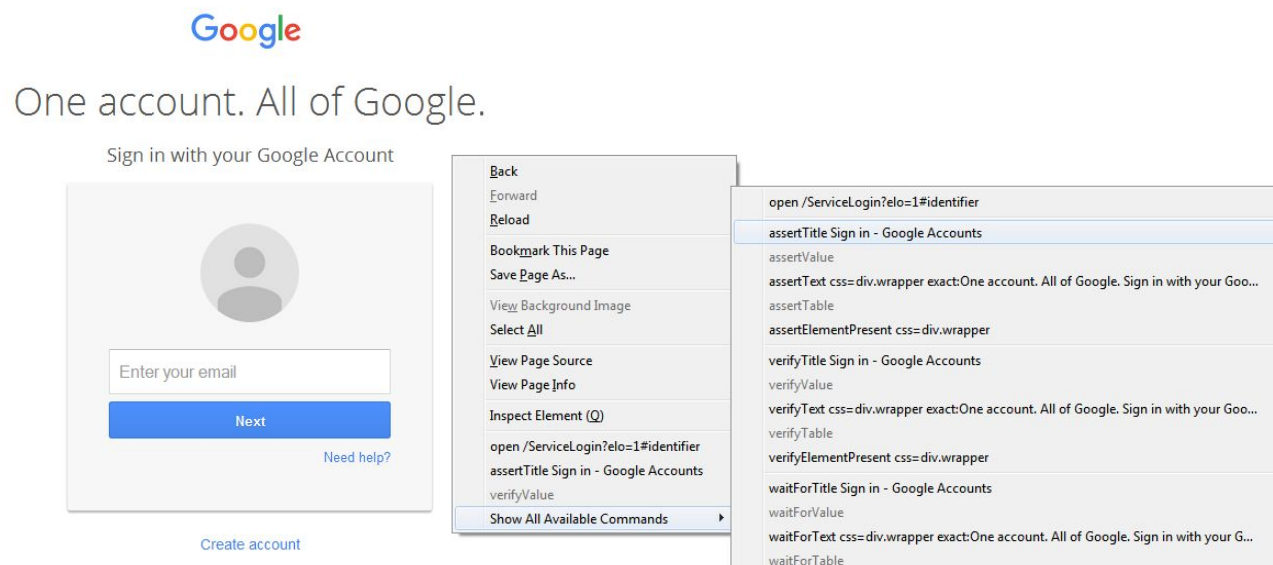
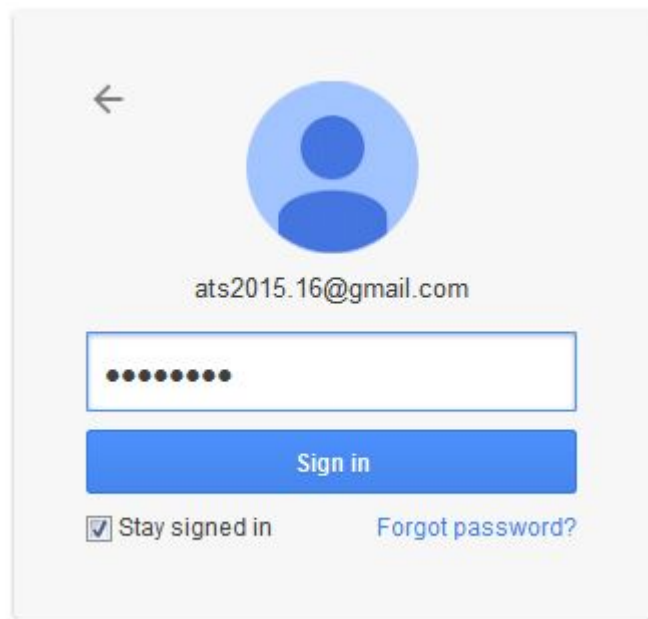


Table	Source	
Command	Target	Value
open	/ServiceLogin?elo=1#identifier	
assertTitle	Sign in - Google Accounts	

Selenium IDE record example - Step 3 & 4

- ▶ Enter email and click next
- ▶ Enter password and click sign in



←



ats2015.16@gmail.com

.....

Sign in

☒ Stay signed in [Forgot password?](#)

Table	Source	
Command	Target	Value
open	/ServiceLogin?elo=1#identifier	
assertTitle	Sign in - Google Accounts	
type	id=Email	ats2015.16@gmail.com
click	id=next	
type	id=Passwd	querias?
clickAndWait	id=signIn	

Selenium IDE record example - Step 5

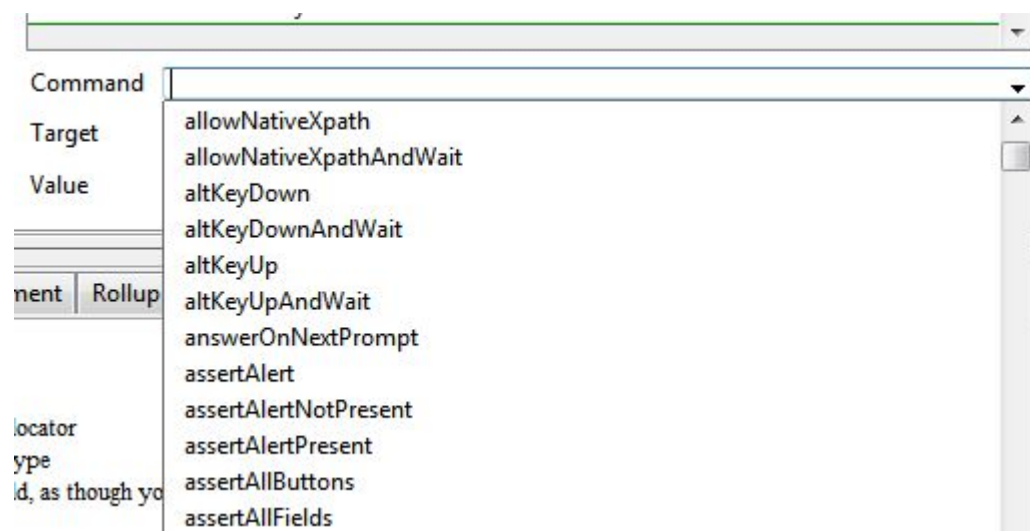
- Right click anywhere not an image or link

The screenshot shows the Selenium IDE interface. On the left, a web page titled "My Account" is displayed with a blue header and a white body containing Google, Gmail, and YouTube icons. A right-click context menu is open over the page, listing various actions like Back, Forward, Reload, and Show All Available Commands. The "Show All Available Commands" option is selected. On the right, the "Commands" panel shows a list of recorded commands. Below this, the "Table" tab is active, displaying a table of commands and their targets.

Command	Target	Value
assertTitle	Sign in - Google Accounts	
type	id=Email	ats2015.16@gmail.com
click	id=next	
assertTitle	My Account	
type	id=Passwd	querias?
clickAndWait	id=next	
assertTitle	My Account	

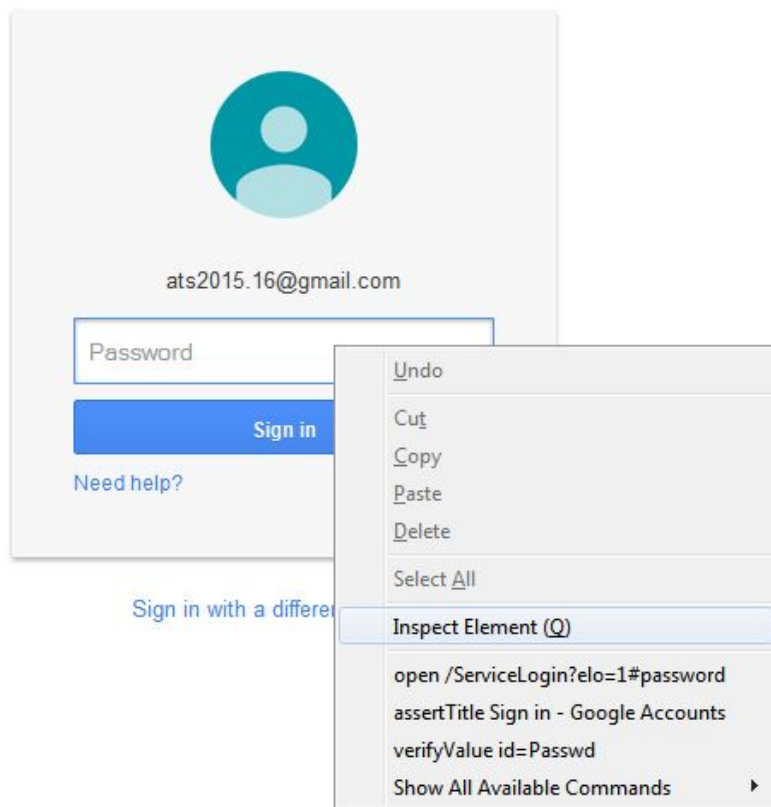
Selenium IDE manual example?

- You can also manually add test steps from a large list of options



Selenium IDE manual example?

- ▶ You can also manually add test steps from a large list of options
- ▶ Right click on an element and press “Inspect Element (Q)”



Selenium IDE manual example?

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- ▶ You can also manually add test steps from a large list of options
- ▶ Right click on an element and press “Inspect Element (Q)”
- ▶ Look at source html for the **element id**

The screenshot displays the Selenium IDE interface with the 'Inspector' tab active. The breadcrumb trail shows the path: html > body > div.wrapper > div.main.content.clearfix > div.card.signin-card > form#gaia_loginform > input#Passwd. The HTML source on the left highlights the password input field: `<input id="Passwd" class="" type="password" placeholder="Password" name="Passwd" ></input>`. The CSS rules on the right show the styles for the selected element, including `margin-top: -1px;` and `position: relative;`.

```
<input id="dnConn" type="hidden" value="" name="dnConn" ></input>
<input id="checkConnection" type="hidden" value="youtube:387:1" name="checkConnection" ></input>
<input id="checkedDomains" type="hidden" value="youtube" name="checkedDomains" ></input>
<span id="reauthEmail" class="reauth-email" > ... </span>
<input id="Email" class="email-input hidden" type="email" readonly="" spellcheck="false" value="ats2015.16@gmail.com" placeholder="Email" name="Email" ></input>
<label class="hidden-label" for="Passwd" > ... </label>
<input id="Passwd" class="" type="password" placeholder="Password" name="Passwd" ></input>
<input id="signIn" class="rc-button rc-button-submit" type="submit" value="Sign in" name="signIn" ></input>
<a id="link-forgot-passwd" href="https://accounts.google.com/RecoverAccount?Email=ats2015.16%40gmail.com" > ... </a>
</form>
</div>
<div class="one-google" > ... </div>
</div>
<div class="google-footer-bar" > ... </div>
</div>
<script > ... </script>
<script type="text/javascript" > ... </script>
<script > ... </script>
<script > ... </script>
<script type="text/javascript" > ... </script>
<script type="text/javascript" > ... </script>
<script > ... </script>
<script > ... </script>
<script > ... </script>
</body>
</html>
```

```
element {
}
.signin-card #Email:hover, inline:716
.signin-card #Email:focus,
.signin-card #Passwd:hover, .signin-card
#Passwd:focus {
  z-index: 3;
}
.signin-card #Passwd { inline:709
  margin-top: -1px;
}
.signin-card #Email, inline:646
.signin-card #Passwd,
.signin-card .captcha {
  direction: ltr;
  height: 44px;
  font-size: 16px;
}
.signin-card inline:633
input[type="email"],
.signin-card input[type="password"],
.signin-card input[type="text"],
.signin-card input[type="submit"] {
  width: 100%;
  display: block;
  margin-bottom: 10px;
  position: relative;
  -moz-box-sizing: border-box;
}
```

Selenium IDE manual example?

- ▶ You can also manually add test steps from a large list of options
- ▶ Right click on an element and press “Inspect Element (Q)”
- ▶ Look at source html for the **element id**

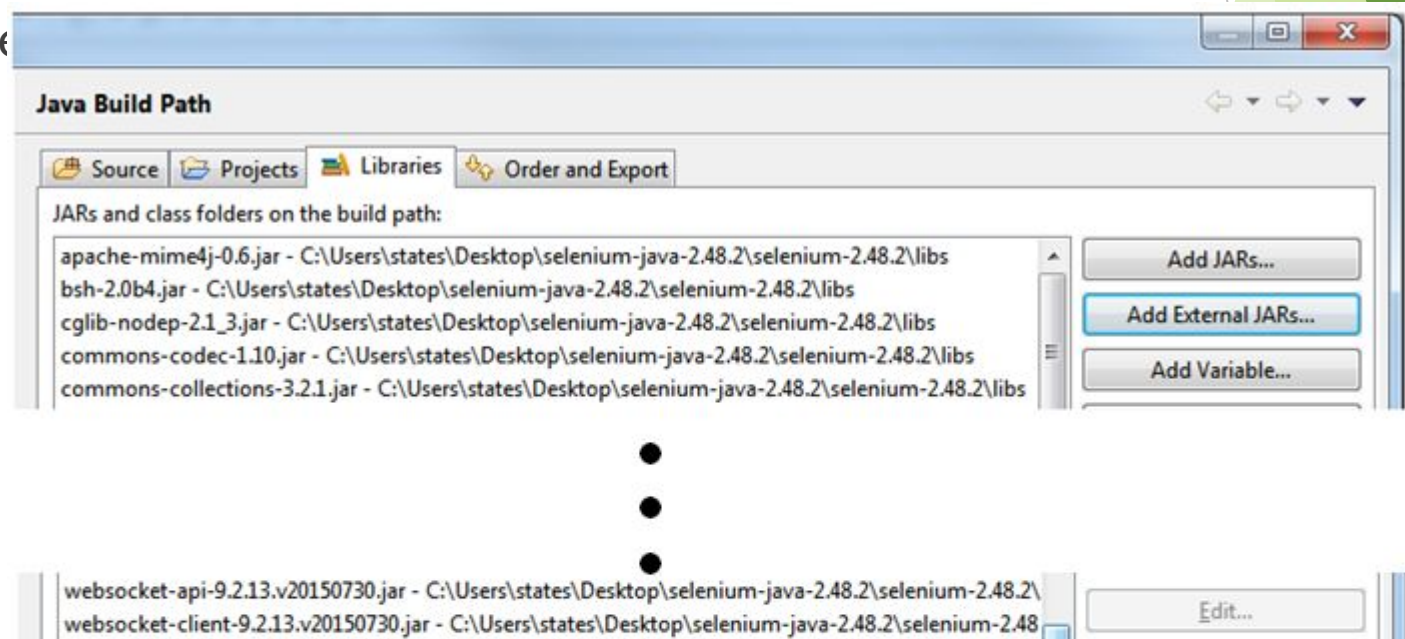
You can also use other source code locators:

- ▶ Firebug
- ▶ Xpath checker
- ▶ CSS Selector
- ▶ Whatever floats your boat :)

WebDriver

WebDriver install

- ▶ Prerequisite Java & Eclipse
- ▶ Step 1 - Download WebDriver (<http://docs.seleniumhq.org/download/>)
- ▶ Step 2 - Extract folder (selenium-java-X.XX.X.zip)
- ▶ Step 3 - Open Eclipse and create new project (File>New>Java Project)
- ▶ Step 4 - Right click on our new project and select **Properties**
- ▶ Step 5 - Java Build Path > Libraries. Click “Add External Jars”
- ▶ Step 6 - Select **ALL** the JAR files



WebDriver example

Create script to:

- ▶ 1 - Launch browser and open “https://accounts.google.com”
- ▶ 2 - Verify title of the page and print verification result
- ▶ 3 - Enter username and password
- ▶ 4 - Click on the Sign in button
- ▶ 5 - Close the browser

WebDriver example - Step 0

- ▶ Create new java class (Gmail_Login.java for example)
- ▶ Import the following

```
import org.openqa.selenium.By;  
import org.openqa.selenium.WebDriver;  
import org.openqa.selenium.WebElement;  
import org.openqa.selenium.firefox.FirefoxDriver;
```

WebDriver example - Step 1

```
// objects and variables instantiation
WebDriver driver = new FirefoxDriver();
String appUrl = "https://accounts.google.com";

// launch the firefox browser and open the application url
driver.get(appUrl);

// maximize the browser window
driver.manage().window().maximize();
```

WebDriver example - Step 2

```
// declare and initialize the variable to store the expected title of the
// webpage.
String expectedTitle = "Sign in - Google Accounts";

// fetch the title of the web page and save it into a string variable
String actualTitle = driver.getTitle();
System.out.println("title:"+actualTitle);

// compare the expected title of the page with the actual title of the page
// and print the result
if (expectedTitle.equals(actualTitle)) {
    System.out.println("Verification Successful - The correct title
                        is displayed on the web page.");
}
else {
    System.out.println("Verification Failed - An incorrect title
                        is displayed on the web page.");
}
```

WebDriver example - Step 3

```
// enter a valid username in the email textbox
WebElement username = driver.findElement(By.id("Email"));
username.clear();
username.sendKeys("TestSelenium");

// Click on next button
WebElement nxtbtn = driver.findElement(By.id("next"));
nxtbtn.click();

Thread.sleep(1000);

// enter a valid password in the password textbox
WebElement password = driver.findElement(By.id("Passwd"));
password.clear();
password.sendKeys("password123");
```

WebDriver example - Step 4 & 5

```
// click on the Sign in button
WebElement SignInButton =
driver.findElement(By.id("signIn"));
SignInButton.click();
```

```
// close the web browser
driver.close();
```

```
System.out.println("Test script executed successfully.");
```

```
// terminate the program
System.exit(0);
```

JUnit?

```
import org.openqa.selenium.By;
import org.openqa.selenium.WebDriver;
import org.openqa.selenium.WebElement;
import org.openqa.selenium.firefox.FirefoxDriver;

public class Gmail_Login{
    @Test
    public void firstTest(){
        // objects and variables instantiation
        WebDriver driver = new FirefoxDriver();
        String appUrl = "https://accounts.google.com";
        .
        .
    }
```

JUnit?

@Test

```
public void firstTest() {  
    .  
    .  
    // compare the expected title of the page with the  
    // actual title of the page and print the result  
    if (expectedTitle.equals(actualTitle)) {  
        System.out.println("Verification Successful - The  
        correct title is displayed on the web page.");  
    }  
    else {  
        System.out.println("Verification Failed - An  
        incorrect title is displayed on the web page.");  
    }  
    .  
    .  
}
```

JUnit?

@Test

```
public void firstTest() {
```

```
    .
```

```
    .
```

```
    Assert.assertEquals("Verification Failed - An incorrect title is  
displayed on the web page.",expectedTitle,actualTitle);
```

```
    .
```

```
    .
```

```
}
```

Helpful API examples

Type	Syntax	Description
id	<code>driver.findElement(By.id("ID_of_Element"))</code>	Locate by value of the "id" attribute
className	<code>driver.findElement(By.className("Class_of_Element"))</code>	Locate by value of the "class" attribute
linkText	<code>driver.findElement(By.linkText("Text"))</code>	Locate by value of the text of the hyperlink
partialLinkText	<code>driver.findElement(By.partialLinkText("PartialText"))</code>	Locate by value of the sub-text of the hyperlink
name	<code>driver.findElement(By.name("Name_of_Element"))</code>	Locate by value of the "name" attribute
xpath	<code>driver.findElement(By.xpath("Xpath"))</code>	Locate by value of the xpath
cssSelector	<code>driver.findElement(By.cssSelector("CSS Selector"))</code>	Locate by value of the CSS selector
tagName	<code>driver.findElement(By.tagName("input"))</code>	Locate by value of its tag name

References

Presentation based on:

- ▶ <http://www.softwaretestinghelp.com/selenium-tutorial-1/>

Selenium

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